

Semiparametric Proportional Hazards Regression — Cheatsheet

BIOST 3050 / STAT 2261: Survival Analysis — Lectures 16 & 17

The Cox PH Model

General semiparametric PH model:

$$h(t | Z) = h_0(t) C(\beta^\top Z)$$

- $h_0(t)$: baseline hazard (nonparametric, unspecified)
- $C(\cdot)$: known, nonneg., $C(0) = 1$
- β : p -dim regression coefficients (parametric)

Cox PH model (canonical $C = \exp$):

$$h(t | Z) = h_0(t) \exp(\beta^\top Z)$$

Data: (T_j, δ_j, Z_j) , $j = 1, \dots, n$, with $T_j = \min(X_j, C_j)$, $\delta_j = \mathbf{1}(X_j \leq C_j)$.

Hazard Ratio & Interpretation

Hazard ratio (free of t):

$$\frac{h(t | Z_1)}{h(t | Z_2)} = \exp[\beta^\top (Z_1 - Z_2)]$$

- $\beta > 0$: hazard $\uparrow$, survival $\downarrow \Rightarrow$ harmful effect.
- $\beta < 0$: hazard $\downarrow$, survival $\uparrow \Rightarrow$ beneficial effect.
- e^β : hazard ratio per one-unit increase in Z .

Coding Covariates

Continuous Coding

Entering a K -level variable as continuous (e.g., stage = 1, 2, 3, 4) forces equally spaced log-HRs: $\exp(\beta)$, $\exp(2\beta)$, $\exp(3\beta)$.

Dummy (Indicator) Coding

Use $K - 1$ dummy variables; reference level has all dummies = 0. Each β_k is log-HR of level k vs. reference (unrestricted spacing).

Example (4 stages, ref = stage 1):

$$h(t | Z) = h_0(t) \exp(\beta_1 Z_1 + \beta_2 Z_2 + \beta_3 Z_3 + \beta_4 \text{age})$$

$Z_k = 1$ if stage $k+1$, else 0.

Interaction Terms

To test whether the age effect differs by stage, add interactions $Z_5 = Z_1 \times \text{age}$, etc. Test $H_0 : \beta_5 = \beta_6 = \beta_7 = 0$.

Factorial Coding (2×2)

Code two main effects + interaction. E.g., for race $\times$ gender: $Z_1 = \text{female}$, $Z_2 = \text{Black}$, $Z_3 = Z_1 Z_2$.

Partial Likelihood (No Ties)

Let $t_1 < \dots < t_D$ be the ordered distinct event times ($d_i = 1$).

$Z_{(i)}$: covariates of subject failing at t_i .

$R(t_i) = \{j : T_j \geq t_i\}$: at-risk set.

Partial likelihood:

$$L(\beta) = \prod_{i=1}^D \frac{\exp(\beta^\top Z_{(i)})}{\sum_{j \in R(t_i)} \exp(\beta^\top Z_j)}$$

Key insight: $h_0(t_i)$ cancels in numerator/denominator.

Log partial likelihood:

$$\ell(\beta) = \sum_{i=1}^D \left[\beta^\top Z_{(i)} - \log \left(\sum_{j \in R(t_i)} e^{\beta^\top Z_j} \right) \right]$$

Rigorous derivation: Profile the full likelihood over h_0 ; the MLE of $h_i = h_0(t_i)$ is

$$\hat{h}_i = \frac{1}{\sum_{j \in R(t_i)} e^{\beta^\top Z_j}}$$

Substituting back yields $L(\beta)$ up to a constant.

Breslow Estimator

Baseline cumulative hazard:

$$\hat{H}_0(t) = \sum_{i: t_i \leq t} \frac{1}{\sum_{j \in R(t_i)} \exp(\hat{\beta}^\top Z_j)}$$

Score Function & Information

Score:

$$U(\beta) = \sum_{i=1}^D \left[Z_{(i)} - \frac{\sum_{j \in R(t_i)} Z_j e^{\beta^\top Z_j}}{\sum_{j \in R(t_i)} e^{\beta^\top Z_j}} \right]$$

Each term = observed covariate – risk-set weighted mean.

Observed information:

$$I(\beta) = \sum_{i=1}^D \left[\frac{\sum_j Z_j Z_j^\top e^{\beta^\top Z_j}}{\sum_j e^{\beta^\top Z_j}} - \left(\frac{\sum_j Z_j e^{\beta^\top Z_j}}{\sum_j e^{\beta^\top Z_j}} \right)^{\otimes 2} \right]$$

where $v^{\otimes 2} = vv^\top$ and sums are over $j \in R(t_i)$.

MLE: $\hat{\beta}$ solves $U(\beta) = 0$ (Newton–Raphson).

Asymptotics: $\hat{\beta} \sim N(\beta_0, I^{-1}(\beta_0))$.

$\widehat{\text{Var}}(\hat{\beta}) = I^{-1}(\hat{\beta})$.

Global Tests for $H_0 : \beta = \beta_0$

All $\sim \chi_p^2$ under H_0 :

1. **Wald:** $\chi_W^2 = (\hat{\beta} - \beta_0)^\top I(\hat{\beta}) (\hat{\beta} - \beta_0)$
2. **LR:** $\chi_{LR}^2 = 2[\ell(\hat{\beta}) - \ell(\beta_0)]$
3. **Score:** $\chi_{SC}^2 = U(\beta_0)^\top I^{-1}(\beta_0) U(\beta_0)$

Score test advantage: Does not require computing $\hat{\beta}$.

Score Test $\equiv$ Log-Rank Test

For a single binary covariate $Z \in \{0, 1\}$:

$$\chi_{SC}^2 = \frac{U(0)^2}{I(0)} = \frac{Z_1(\tau)^2}{\hat{\sigma}_{11}(\tau)}$$

which is exactly the two-sample log-rank statistic. The log-rank test is a special case of the Cox PH framework.

Partial Likelihood with Ties

At each t_i : $d_i \geq 1$ failures; $D(t_i)$ = failure set; $S_i = \sum_{j \in D(t_i)} Z_j$.

1. Breslow (default in SAS)

$$L_1(\beta) = \prod_{i=1}^D \frac{\exp(\beta^\top S_i)}{\left[\sum_{j \in R(t_i)} e^{\beta^\top Z_j}\right]^{d_i}}$$

Simple; poor with heavy ties.

2. Efron (default in R)

$$L_2(\beta) = \prod_{i=1}^D \frac{e^{\beta^\top S_i}}{\prod_{k=1}^{d_i} \left[\sum_{j \in R(t_i)} e^{\beta^\top Z_j} - \frac{k-1}{d_i} \sum_{j \in D(t_i)} e^{\beta^\top Z_j} \right]}$$

Better approximation than Breslow; progressively removes tied failures from denominator.

3. Cox Exact (discrete model)

$$L_3(\beta) = \prod_{i=1}^D \frac{\exp(\beta^\top S_i)}{\sum_{q \in \mathcal{Q}_i} \exp(\beta^\top S_q^*)}$$

$\mathcal{Q}_i$: all $\binom{|R(t_i)|}{d_i}$ subsets of size d_i from $R(t_i)$. Exact but computationally expensive.

	Breslow	Efron	Exact
Accuracy	Low	Mid	Best
Speed	Fast	Fast	Slow
R default		✓	

When $d_i = 1 \forall i$, all three reduce to the standard PL.

Local Tests for $H_0 : \beta_1 = \beta_{10}$

- Partition: $\beta = (\beta_1, \beta_2)^\top$, β_1 is q -dim.
 $I^{11}(\beta)$ = upper-left $q \times q$ block of $I^{-1}(\beta)$.
1. **Wald:** $(\hat{\beta}_1 - \beta_{10})^\top [I^{11}(\hat{\beta})]^{-1} (\hat{\beta}_1 - \beta_{10})$
 2. **LR:** $2[\ell(\hat{\beta}) - \ell(\beta_{10}, \hat{\beta}_2(\beta_{10}))]$
 3. **Score:** $U_1^\top I^{11} U_1$ (evaluated at restricted MLE)
- All $\sim \chi_q^2$ under H_0 .

Linear Combination Test

Test $H_0 : C\beta = C\beta_0$ where C is $q \times p$ full-rank:

$$\chi_W^2 = (C\hat{\beta} - C\beta_0)^\top [C I^{-1}(\hat{\beta}) C^\top]^{-1} (C\hat{\beta} - C\beta_0)$$

Example: $H_0 : \beta_1 = \beta_2 = \beta_3$ with $\beta = (\beta_1, \beta_2, \beta_3, \beta_4)^\top$:

$$C = \begin{pmatrix} 1 & -1 & 0 & 0 \\ 0 & 1 & -1 & 0 \end{pmatrix}$$

R Software: `coxph()`

```
library(survival)
fit <- coxph(Surv(time, status) ~ Z1 + Z2,
             data = dat,
             ties = "efron") # or "breslow", "exact"
summary(fit)
# Reports: coef, exp(coef), se, z, p
# Global: LR, Wald, Score tests
```

`summary(fit)` gives all three global tests automatically.